

Neural Network-based Classification of ENG Recordings in Response to Naturally Evoked Stimulation

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ABSTRACT

The paper evaluates two different neural network strategies for the classification of electroneurographic (ENG) recordings obtained in response to mechanical stimulations of a rat paw available from an open-access dataset. The first strategy is based on spiking neural networks (SNNs), a new class of artificial neural networks inspired by the information processing solutions of biological neurons. SNNs are considered to be promising for the classification of complex space-time models. The second strategy relies on convolutional neural networks (CNNs), a well-established tool for processing structured arrays of data such as images or space-time data. The existing effects of power line noise and other distortions generated by the non-idealities of the acquisition system (e.g., timing jitter) are removed by a suitable pre-processing. The accuracy and F1-score achieved by the SNN and the CNN are reported for a multi-class problem aiming to separate different types of somatosensory stimuli (i.e., nociception, plantar flexion, dorsiflexion, and touch, which are the mechanical stimulations of the dataset) and their different intensities.

CCS CONCEPTS

• **Information systems** → *Spatial-temporal systems*; • **Applied computing** → *Bioinformatics*.

KEYWORDS

Electroneurogram, neural networks, nerve cuff electrode, rat sciatic nerve, signal preprocessing and classification

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1 BACKGROUND

Peripheral neuropathies are an increasingly important clinical case to be considered. Because the number of affected people is destined to increase over time, in order to tackle the problems linked with such a disease, it will be necessary to study and propose alternative solutions to traditional medicine approaches. In fact, peripheral nerve injuries have a serious impact on the daily life of patients as they induce sensory and motor deficits in the areas of the body innervated by the affected nerve, which often leads to total or partial paralysis of the district [22]. Nerve regeneration surgery has shown that only about 40% achieve satisfactory motor and sensory recovery [18]. Clinical and preclinical advances in biological and cellular strategies for reconstruction are not yet adequate to support functional recovery [18]. Thus, new strategies focusing on innovative technologies are needed.

In recent years, scientific and technological efforts have been made towards the development of neuroprostheses with the aim of bypassing the injury site and restoring the flow of information both to and from the central nervous system [15, 20]. Although interesting results have been achieved in the open-loop control of lower and upper limb movement [1, 7], the incorporation of sensory neural feedback into prosthetic devices is still a challenge.

A schematic of a possible implanted neuroprosthetic device for the joint recovery of the hand is shown in Fig. 1. The implanted device samples the peripheral nerve signal - the electroneurogram (ENG) - in the around of the injured area, or even from other useful areas. The sampled signal is then transmitted to a data processing

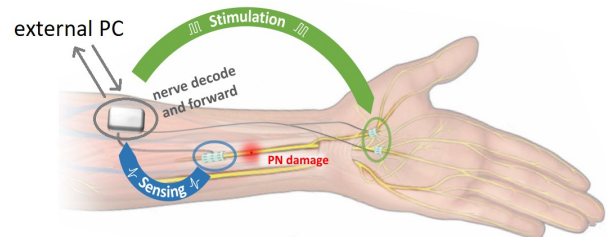


Figure 1: Schematic example of a fully implanted device for hand functional recovery.

unit, which can be external (e.g., an external PC) or embedded in the implanted circuitry. After decoding the information recorded, the

nerve can be artificially stimulated to replicate the neural response of the body.

Peripheral neural interfaces based on nerve cuff electrodes (e.g., extraneural electrodes wrapped around the outer surface of the nerve) are often considered the preferable solution to measure the ENG signal. In fact, this type of interfaces demonstrated the long-term stability necessary for clinical applications in humans [19]. However, extraneural electrodes typically suffer from a lower signal-to-noise ratio (SNR) with respect to their more invasive counterparts, i.e. intraneural electrodes [11]. Therefore, the decoding of sensory information recorded by a nerve is a non-trivial task and a well-established technique is yet to be found. In this context, the contributions of this work are two-folds:

- Since the ENG with a very low-amplitude (μV) is often affected by powerline interference (PLI) and other distortions generated by non-idealities of the acquisition system, a preprocessing strategy that is able to facilitate the development of any downstream classifier has been delineated. Further, we have proposed a novel signal regridding method that exploits the presence of PLI for correcting the timing distortions caused by the timing jitter of the acquisition system.
- Two different classification strategies to decode naturally evoked ENG responses have been tuned to the use case at hand. The first one is based on spiking neural networks (SNNs), a novel class of artificial neural networks inspired by the information processing solutions of the biological neurons that seems to be very promising for the classification of temporal patterns in stochastic signals (note that whole-nerve recordings can be modelled as Gaussian processes [9]). The second one is based on convolutional neural networks (CNNs), which are the state-of-the-art classifiers and thus set the background performance method.

The reminder of this paper is organized as follows. In Section 2 we briefly discuss the main classification strategies found in literature for ENG signals recorded via extraneural cuff electrodes. In Section 3 we describe the public dataset that was leveraged in this analysis. In Section 4 we present the signal preprocessing steps that also comprise the novel signal regridding strategy. In Section 5 we describe the SNN and CNN classification approaches. In Section 6 we present and discuss the achieved results, and finally, in Section 7, conclusions are drawn.

2 RELATED WORKS

Several techniques to classify naturally evoked ENG activity have been explored, but a well-suited mode of classification has yet to be established and the field is still open to exploration. Depending on the contacts configuration of the sensing interface, different signal analysis strategies were exploited. For extra-neural cuff electrodes, it is possible to define the following families of electrode configurations:

Single-Channel Cuff Electrode has from 1 to 3 electrodes, depending on the reading mode (i.e., tripolar, bipolar, or monopolar), and performs the measurement on a single channel [11]. Electrodes are ring-shaped and encircle the perimeter of the nerve (Fig. 2a). In this case, the main signal processing techniques rely on extracting statistical information from the recorded ENG. For example, one can infer the firing frequency of the family of active neurons within the

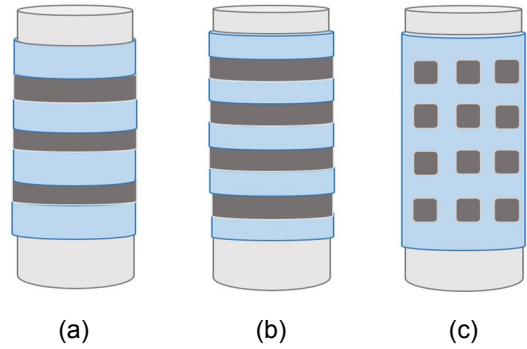


Figure 2: Different contacts configuration for an extraneural cuff electrode: (a) single-channel cuff electrode; (b) multi-ring cuff electrode; (c) multi-contact cuff electrode.

nerve. The simplest approach relies on the use of a logical threshold to detect the compound action potentials (CAPs), i.e. the signature activity resulting from the simultaneous activation of a given family of neurons. The firing rate is defined as the number of times a CAP is observed over time. The concept underlying the model is that there is a temporal delay between consecutive CAPs to which a probability of occurrence of the event can be associated. Then, the ENG can be classified both based on the shape of the detected CAPs, and on the firing rate. Alternatively, a running observation window (ROW) approach can be used to extract information from the ENG. In this method, a rectangular window of fixed time length is shifted over the recorded ENG signals. A set of features (e.g., standard deviation, mean absolute value, etc.) is then computed over the data contained in each window and used to train a machine learning classifier. Raspovic *et al.* [17], for example, extracted both time-domain and frequency-domain features from the windows and used support vector machine (SVM) and artificial neural network (ANN) classifiers.

Multi-Ring Cuff Electrode is composed of several ring electrodes, typically more than 3 (Fig. 2b) [21]. In this way, it is possible to have information on several channels along the longitudinal direction. If the electrodes are far enough one another, one can extract information related to the conduction velocity of the signal. Since the conduction velocity is different among fiber types, its estimate enables to infer the type of fiber that is active within the nerve. Velocity Selective Recordings (VSR) algorithms can be used to extract velocity. To do so, a process named delay-and-add is commonly used [24]. This approach lays its foundations on the fact that the signal that arrives at consecutive rings is the "same" if we account for a given delay, minus the noise. Therefore, estimating the delay allows to get an estimate of the propagation velocity. Example of VSR applied to naturally evoked ENG recordings are found in [13, 14]. Of course, the ROW approach can be used with multi-ring cuff electrodes as well.

Multi-Contact Cuff Electrode has multiple rings, and each ring presents $M > 1$ individual contacts (Fig. 2c) [10]. The total number of contacts amounts to $M \times N$, if N is the number of rings. Thus, the geometry of the electrodes is matrix shaped. In this way, a certain

level of spatial resolution (i.e., spatial information related to the angular position over the ring(s) of the source that generated the ENG signal inside the nerve) is also provided, improving ENG selectivity. This information can be exploited by source localization or beam-forming approaches [23, 26], which allow to estimate and separate the activity originated from a given location in the transversal section of the nerve. Multi-contact cuffs also allow to combine spatial and temporal information. An example is provided by [10], where different classifiers (such as matching filters, ANNs and CNNs) were used to associate single CAPs with the different sensory fibers that generated them. The CAPs were identified in the ENG signal through a thresholding logic, and then their spatio-temporal representation (i.e., an $M \times T$ matrix containing the information extracted from M contacts at T consecutive time samples) was fed to the classifiers. Finally, Brunton *et al.* in [3] applied the ROW-based approach to the multi-contact electrode configuration and found that neural responses were separable both based on somatosensory stimulus type (i.e., proprioception, nociception, touch) and on the intensity of the stimuli.

3 THE OPEN-SOURCE ENG DATASET

In the analysis herein, the classification pipeline was developed and tested using the open-source ENG dataset from the Newcastle University [2], which is described in [3]. The dataset includes ENG recordings in response to mechanical sensory stimulation, which were acquired through a multi-contact extra-neural cuff electrode (16 contacts, with 4 rings of 4 platinum contacts each, Fig. 3) placed on the sciatic nerve of three different rats. Ten different classes of stimuli were administered to the hind paw of the rats, grouped into four main stimulus etiologies:

- *Nociception*: two classes consisting of two different pinch positions on the foot, which are the pinch of the heel and the external pinch of the toe;
- *Negative proprioception*: three different angles for plantar flexion of the ankle (-10° , -20° , -30° with respect to the resting position of the ankle);

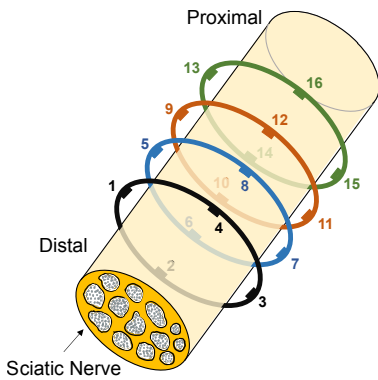


Figure 3: Schematic of the 16-contact cuff electrode used in [2] to acquire the ENG signals in analysis.

- *Positive proprioception*: three different angles for dorsiflexion of the ankle ($+10^\circ$, $+20^\circ$, $+30^\circ$ with respect to the resting position of the ankle);
- *Touch*: two classes consisting of 0.1 kg touch and 0.3 kg touch.

Fifty repetitions were performed for each stimulus class, for a total of 500 samples (10 classes $\times$ 50 samples) available for each rat. The data was collected with a sampling rate of 30 kHz.

4 ENG PREPROCESSING

Before providing ENG samples to the classifiers, the signals were carefully preprocessed to filter out noise sources and select the frequencies of interest. Two main sources of noise were encountered:

- The PLI, which was heavily overlaid on all recordings. Although the lower harmonics were the most powerful, some harmonic frequencies were superimposed on the ENG bandwidth (i.e., 0.8–2.5 kHz) up to 1.5 kHz.
- The distortions related to the sampling jitter, which were highly visible in the representation of the time frequency of the signal.

Fig. 4b shows the short-time Fourier transform (STFT) of one of the signals at one of the harmonics of PLI at 650 Hz. The distortions are caused by the timing jitter in the time domain that are interpreted as time-dependent changes of the fundamental frequency of the periodic signals. In this case, the sampling errors were not negligible and any notch filter applied to remove the PLI would be ineffective for this reason. It is worth to highlight that any distortion would severely impair the performance of the downstream classifier. In fact, the PLI can be variable depending on the instrumentation used for a given stimulation class, and since all the samples belonging to a class have been acquired consecutively, the classifier trained on such data can learn to recognize the specific harmonic content of the powerline that overlaps with a specific label instead of extracting meaningful, informative ENG content. The preprocessing pipeline implemented in this work consisted of 4 phases:

- (1) *Regridding*: we derived a new methodology to correct the irregular time grid of the signals and obtain a uniform one (i.e., signal regridding). This was done starting from the STFT of the signals and exploiting the presence of PLI in this domain. In particular, by quantifying the displacement of the PLI in the time-frequency domain with respect to time, it is possible to estimate the instants in which the signal was really sampled. The following relationship can be obtained (Fig. 4a) for frequency deviation

$$\Delta f_s(t) \simeq -f_{so} \times \frac{\Delta \hat{f}_0(t)}{f_0}, \quad (1)$$

where f_{so} is the nominal sampling frequency (30 kHz in our case), f_0 is the real frequency of a PLI harmonic chosen to estimate $\Delta \hat{f}_0(t)$ (in our case $f_0 = 750$ Hz), $\Delta \hat{f}_0(t)$ is the estimate of the time shift (with respect to f_0) of the peak frequency of the harmonic observed in the time-frequency domain (linear interpolation of the ceiling points around 750 Hz was used in this work), and $f_s(t) = \frac{1}{T_s(t)}$ is the reciprocal of the time-varying sampling interval $T_s(t)$ (i.e., the real instantaneous sampling frequency). Then, the instants of the real sampling

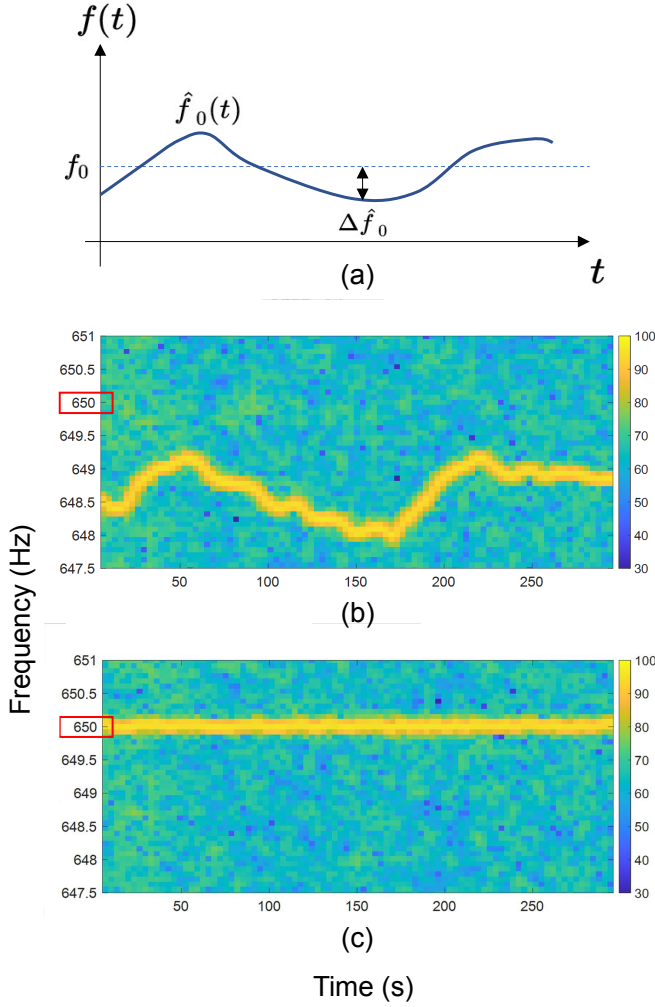


Figure 4: (a) Representation of the frequency variables in the regridding problem. (b) STFT of one of the signals in the dataset zoomed in the proximity of 650 Hz before the regridding procedure. (c) STFT of the same signal as in (b) after the regridding procedure was applied.

time t_i (with $i = 0, 1, 2, \dots$) can be calculated iteratively as:

$$\begin{cases} t_0 = 0, \\ \hat{t}_k = t_0 + \sum_{i=1}^k f_s^{-1}(t_i). \end{cases} \quad (2)$$

Once the instants in which the samples were really drawn are known, one can interpolate this non-uniform time grid to reach a uniform one (spline interpolation was used in this work).

- (2) *PLI cancellation*: PLI harmonics have been removed by applying Widrow's adaptive notch filter [16]. In fact, thanks to its recursive nature, it can efficiently circumvent any residual sampling jitter effect that we could not cancel through the regridding process. Furthermore, its high selectivity, frequency tracking

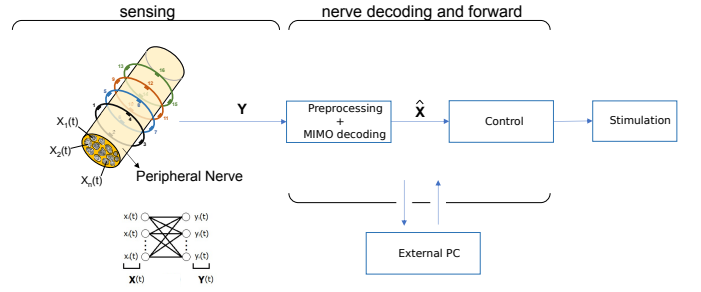


Figure 5: Information flow for MIMO nerve decoding.

capability, low distortion, and low computational cost characteristics are particularly suitable for bioelectrical acquisition systems.

- (3) *Bandpass Filtering*: to extract the ENG frequencies of interest, an 8th order Butterworth bandpass filter between 0.8 kHz and 2.5 kHz was applied bidirectionally (for zero-phase filtering).
- (4) *Downsampling*: the signals were finally down-sampled at 5600 Hz to reduce as much as possible the computational load in the downstream classification phase while maintaining all the information contained in the ENG bandwidth.

5 CLASSIFICATION

Redundant measurement by an implanted cuff with multiple contacts behaves like a multi-input and multi-output (MIMO) system (Fig. 5) in which each (n th) bundle has the signal $x_n(t)$ and each of the cuff contacts measures $y_k(t)$ as a complex and non-linear combination of $\{x_1(t), x_2(t), \dots, x_N(t)\}$. The goal of neural network-based ENG decoding is to (implicitly) separate the different sensory information coming from different family of nerve fascicles.

A SNN and a CNN have been trained to classify the 10 classes of somatosensory stimuli described in Section 3. In both cases, we fed to the network windows of signal of length 1 s. The size was decided to find a fair compromise between computational cost for training by reducing the size of the window and information content stored by increasing its size. As done by other groups [3], we have trained a separate classifier for each animal. To estimate classification performance, we used 10-fold cross-validation with layered distributions. To understand how the preprocessing steps affected the classification performance, we trained and tested the two types of neural networks both on preprocessed signals (steps 1 to 4 described in Section 4 were applied) and on signals where only the bandpass filter and downsampling were applied (only steps 3 and 4, without PLI cancellation and regridding). To allow comparisons with other works and to study the classifiers results, we post-processed the results of the 10-class discrimination problem to find out what the performance would have been on a problem having only 4 classes, depicting the 4 stimulus etiologies described in Section 3: nociception, plantar flexion, dorsi flexion, and touch. To do this, the different stimulation modalities belonging to the same type of sensory stimulation have been aggregated: heel pinch and outer toe pinch were aggregated into forming the general nociception class, 0.1 kg and 0.3 kg of touch were aggregated into the general touch class, ankle positions of -10° , -20° , -30° were

aggregated into the general plantar flexion class, and ankle positions of $+10^\circ$, $+20^\circ$, $+30^\circ$ were aggregated into the general dorsiflexion class. [5].

5.1 Spiking neural network

The biological brain's capabilities of capturing complex spatio-temporal patterns, in general, comes from the complex dynamics of networks of spiking neurons and from the plasticity of the synapses connecting them. Because SNNs try to replicate these mechanisms, they are thought to be particularly suitable for the classification of spatio-temporal data [5]. Besides, as inputs are encoded into sparse temporal binary sequences, the implementation of SNNs on embedded systems is more energy-efficient with respect to a regular ANN having the same number of neurons [4].

In an SNN, artificial neurons communicate through sequences of spikes (i.e., temporal sequences of binary values, as opposed to continuous values as in traditional ANNs) (Fig. 6a). As in natural neurons, the SNN operational model has an internal state variable (i.e., the membrane potential) whose value depends on both current and past inputs and on its own history. A spiking artificial neuron performs a spatiotemporal summation of the inputs (Fig. 6b), then, when the membrane potential reaches a threshold, the neuron fires a spike that travels to other neurons in the next layer. Furthermore, whenever an output spike is produced, the internal state of the neuron is restored to its resting value. The building blocks of an SNN model are three:

- (1) Since spiking neurons require spiking inputs, an encoding unit is needed to transform continuous signals into spike trains before feeding them to the network. In this work, the signals were encoded into spike trains by feeding them to a population (i.e., a layer) of current-based (CUBA) integrated-and-fire (IF) neurons as time-varying injected currents [5].
- (2) Then, the computational model of the spiking neuron must be selected: in this work, the widely used Spike-Response Model (SRM) has been adopted. In an SRM, the equation describing the internal state $u_i(t)$ of the i th neuron accounts for the reset mechanism and refractoriness after firing (phenomena observed in natural neurons) (Fig. 6b):

$$u_i(t) = -V_{th} \sum_s \eta(t - t_i^s) + \sum_j \sum_f w_{ji} K(t - t_j^f), \quad (3)$$

where j is the input synapse index, w_{ji} weights the efficacy of the j -th synapse, $K(t)$ is a kernel describing the impulse response of the system (i.e., the response to an input spike), t_j^f are the times of firing of the j th presynaptic neuron, $\eta(t)$ is a kernel describing the response of the system to its own output (i.e., its output spikes), t_i^s are the times of firing of these output spikes, and V_{th} is the threshold over which a spike is produced. We followed the SNN implementation of the model described in [5], where the SRM was reformulated to allow recursive neuronal status updates. In particular, we used the two following exponential kernels:

$$K(t) = V_0 \left(e^{-\frac{t}{\tau_m}} - e^{-\frac{t}{\tau_s}} \right), \quad (4)$$

$$\eta(t) = e^{-\frac{t}{\tau_r}}, \quad (5)$$

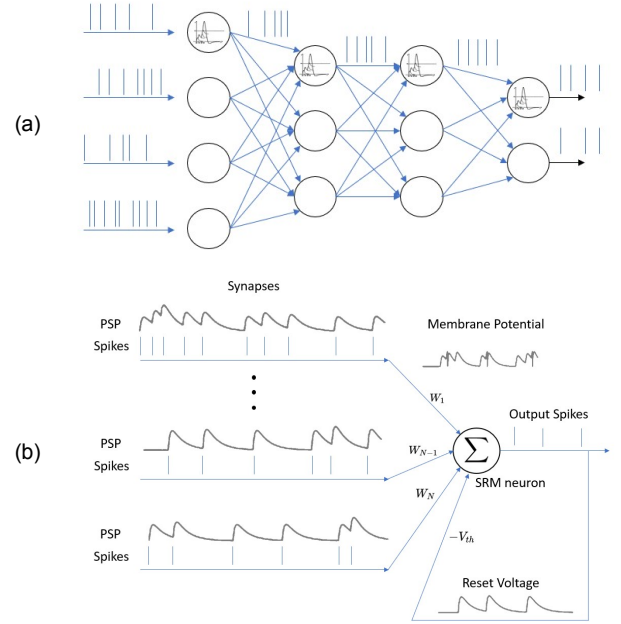


Figure 6: (a) Schematic of the architecture of a SNN: information is exchanged through spike trains, while the mathematical model of the neuron embeds an internal state variable (i.e., the membrane potential) that is updated at each time-step based on the spatiotemporal summation of the inputs. (b) Schematic of a SRM neuron.

where τ_s is the time constant governing the depolarization of the membrane, τ_m is the time constant governing the repolarization after a charge induced by an input spike, and τ_r is the time constant governing the refractoriness of the neuron after an output spike is generated. Normalization of $K(t)$ to unit amplitude was obtained by setting $V_0 = \alpha^{(\alpha-1)} / (\alpha - 1)$ with $\alpha = \tau_m / \tau_s$ [8]. Because it was shown that SNNs made of neurons with learnable time constants are more robust to initial values and can learn faster than SNNs made of neurons with a fixed time constant [6], in our SNN architecture, τ_m , τ_s , and τ_r were initialized manually (respectively at 4, 1, 4) and then adjusted through backpropagation. The SNN architecture used in our work was composed of four layers of respectively 75, 50, 25, 10 SRM neurons.

- (3) Finally, an optimization algorithm must be used to train the network parameters. Since SNNs operate with non-differentiable thresholds, the gradient descent rule cannot be applied directly. To address this, we used a surrogate gradient descent rule in which the derivative of the hard threshold is replaced, in the backward pass, with a smooth approximation, as explained in [5]. Also, since the signals are fed to an SNN in an on-line fashion, we backpropagated the errors using the truncated backpropagation-through time (TBPTT) algorithm were used. We backpropagated the errors every 150 time-steps. Finally, the output class was encoded by the neuron that fired the most, thus, the cross-entropy loss between the one-hot-coded true

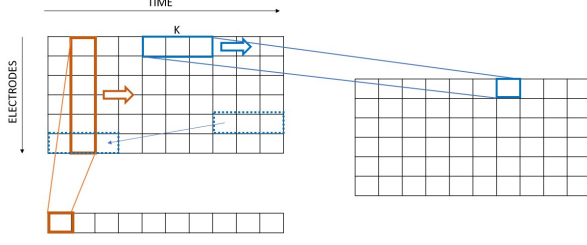


Figure 7: Schematic of the two kernels we used in our 1D convolutional block. The orange kernel’s dimension is $E \times 1$, where E is the width of the data structure along the “electrode dimension”. The blue kernel’s dimension is $1 \times K$, where K is the number of time-steps one wants to convolve to extract local time relationships. Both kernels slide along time only. The output of the first convolution is summed to each row of the output of the second convolution.

label and the average activation rates of the output neurons was used as an optimization criterion.

5.2 Convolutional neural network

The CNN architecture implemented in this work was composed of a combination of 1D convolutional layers. In particular, we designed a 1D convolutional block comprising two types of kernels whose results were then added line by line along the time dimension (Fig. 7). One kernel had the function of carrying out convolutions along the electrodes (orange kernel in Fig. 7), while the other convoluted data only on short sequences along the time dimension (blue kernel in Fig. 7). Each of the kernels was moved with a step length of 1 and zero padding was applied to maintain the same dimensionality after convolution. After four of these custom 1D convolutional blocks, the data was flattened and two more simple 1D convolutions were performed. Finally, the features extracted from the last convolutional layer were fed to a downstream fully connected layer that performs the classification, and the predicted class was one-hot-encoded by the output neurons. A rectified linear unit (ReLU) activation function and a max-pool layer were used between each convolutional layer to respectively add non-linearity and reduce dimensionality. At the exit of the fully connected layer, SoftMax was used to transform the unbounded outputs into probabilities and use them to calculate the cross-entropy loss.

6 RESULTS AND DISCUSSION

Figure 4c shows the effectiveness of the proposed new regriding strategy: the $\Delta f_0(t)$ observed on the raw data is no longer visible in the regrided data. The result is in line with what one would expect from the STFT of a sinusoidal component sampled at regular intervals. After regriding, PLI deletion was made possible and the adaptive notch filter could be applied successfully.

Table 1 shows the mean classification accuracy and the corresponding mean F1 score obtained from the 10-class discrimination problem using pre-processed samples. We can observe that the CNN outperformed the SNN. The CNN’s performance proved to

Table 1: Performance results relative to the 10-classes discrimination problem obtained after preprocessing the signals. Values are reported as *mean $\pm$ standard deviation* calculated over the 10 folds of the cross-validation procedure.

Classifier	Accuracy (%)			F1-score (%)		
	Rat 1	Rat 2	Rat 3	Rat 1	Rat 2	Rat 3
CNN	86.4 $\pm$ 3.6	83.9 $\pm$ 8.7	80.2 $\pm$ 5.1	87.6 $\pm$ 2.9	85.0 $\pm$ 8.1	81.3 $\pm$ 5.4
SNN - CUBA IF encoding	77.2 $\pm$ 5.7	83.7 $\pm$ 5.8	67.8 $\pm$ 8.0	76.8 $\pm$ 6.1	83.7 $\pm$ 6.6	64.7 $\pm$ 8.4

Table 2: Performance results relative to 10-classes discrimination problem with no preprocessing applied (the raw data were only bandpassed and downsampled). Values are expressed as *mean $\pm$ standard deviation* calculated over the 10 folds of the cross-validation procedure.

Classifier	Accuracy (%)			F1-score (%)		
	Rat 1	Rat 2	Rat 3	Rat 1	Rat 2	Rat 3
CNN	81.9 $\pm$ 5.9	85.6 $\pm$ 4.7	81.9 $\pm$ 7.2	83.8 $\pm$ 5.4	87.2 $\pm$ 4.5	83.8 $\pm$ 6.9
SNN - CUBA IF encoding	72.0 $\pm$ 4.5	82.9 $\pm$ 2.8	64.1 $\pm$ 3.4	71.8 $\pm$ 4.8	82.8 $\pm$ 4.1	61.5 $\pm$ 3.0

Table 3: Performance results relative to the 4-classes discrimination problem (i.e., *touch*, *plantiflexion*, *dorsiflexion*, *nociception*) obtained by postprocessing the results of the 10-classes problem. Values are expressed as *mean $\pm$ standard deviation* calculated over the 10 folds of the cross-validation procedure.

Classifier	Accuracy (%)			F1-score (%)		
	Rat 1	Rat 2	Rat 3	Rat 1	Rat 2	Rat 3
CNN	97.1 $\pm$ 1.6	85.3 $\pm$ 7.2	89.5 $\pm$ 6.5	96.9 $\pm$ 1.7	83.8 $\pm$ 7.6	89.1 $\pm$ 7.1
SNN - CUBA IF encoding	96.5 $\pm$ 2.7	88.9 $\pm$ 3.7	86.4 $\pm$ 5.7	96.3 $\pm$ 2.9	87.8 $\pm$ 3.4	84.7 $\pm$ 7.8

be in line with the state of the art: the ROW-based approach by [3] applied on the same dataset used in this work, for example, obtained an accuracy of 0.83 ± 0.03 , 0.84 ± 0.03 , 0.75 ± 0.04 on the three rats respectively (F1 not mentioned). On the other hand, the SNN in this work was slightly inferior than the state of the art performance even if, in the literature, SNNs are presented as a new method particularly suitable for classifying space-time data thanks to plausibility of their internal model [5]. This worse performance can be due to a number of factors: 1- SNN training has emerged in the literature as the main factor limiting their usability [25]. In this work, TBPTT training was found to be computationally burdensome, and the length of time truncation could not be optimized. However, this is a hyperparameter that has a high impact on how much a network can learn. Therefore, a thorough study is needed to understand when to truncate temporal retro propagation while dealing with ENG signals. 2- The spike encoding strategy impacts how much information is retained in the spike train and how noise is rejected/kept, thus, it has the potential to impact classification performance as well. It is possible that there is a more suitable algorithm for encoding useful information in ENG signals into spike sequences. Therefore, this aspect must be studied in future works.

Table 2 shows the classification results of the CNN and the SNN when the signals were not preprocessed and, therefore, PLI and sampling jitter were not eliminated. The performances obtained by CNN were comparable to the “preprocessed” case, while the preprocessing strategy implemented proved useful to improve the

classification performance of the SNN: in fact, an average increase of $\sim 3\%$ was found in both accuracy and in the network's F-1 score. The SNNs could be excellent for investigation in the case of a clean signal, while a high summed noise is seen as a useful signal by the classifier and therefore the information could be misunderstood due to its architecture (the use of a threshold to send a spike). CNN, on the other hand, can recognize such noises and discard them, making the network robust to noise. Table 3 reports the performance related to the 4-classes reduced problem obtained after post-processing the results of the 10-classes one, as explained in Section 5. Both the SNN and the CNN classifiers have shown to perform better in this simplified problem. Importantly, the SNN's performance was in line with CNN's in this case: since the SNN had considerably fewer trainable parameters than CNN (6855 versus 82348), this result could reveal the increased computational power of spiking neurons (with respect to traditional ANN neurons) which was first theorized in [12].

7 CONCLUSION

In this work, we first focused on the preprocessing of ENG signals with the aim of preparing them for the development and testing of a downstream classifier. Secondly, the feasibility of applying SNNs for the classification of naturally evoked afferent ENG activity was demonstrated comparing the results with those that can be obtained from a CNN, which is a more established tool for the classification of space-time models. Although the SNN was slightly inferior to the CNN on a 10-class discrimination problem, the performance of the two instruments was comparable when the problem was reduced to one of 4 classes by merging labels belonging to the same sensory type. Since the SNN used considerably fewer parameters than the CNN, this result could suggest greater efficiency of the spiking computational units. Therefore, particularly in the context of neural prostheses, where limited computational and power resources may be available, we can conclude that SNNs are worth continuing to explore and further develop. Furthermore, this work has individuated some crucial aspects of SNNs that need to be addressed in future works: first, the influence of the length of the truncation of temporal backpropagation has to be studied in details; second, because the encoding method may have a considerable influence on the performance results, it is worth exploring other encoding strategies to find out which is the best suited to encode ENG signal's characteristics.

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